

# 3 PROBABILITY TOPICS

**Figure 3.1** Meteor showers are rare, but the probability of them occurring can be calculated. (credit: modification of work "A meteor during the peak of the 2009 Leonid Meteor Shower" by Navicore/ Wikimedia Commons, CC BY 3.0)

## Chapter Objectives

By the end of this chapter, the student should be able to:

- Understand and use the terminology of probability.
- Determine whether two events are mutually exclusive and whether two events are independent.
- Calculate probabilities using the Addition Rules and Multiplication Rules.
- Construct and interpret Contingency Tables.
- Construct and interpret Venn Diagrams.
- Construct and interpret Tree Diagrams.

## Introduction

It is often necessary to "guess" about the outcome of an event in order to make a decision. Politicians study polls to guess their likelihood of winning an election. Teachers choose a particular course of study based on what they think students can comprehend. Doctors choose the treatments needed for various diseases based on their assessment of likely results. You may have visited a casino where people play games chosen because of the belief that the likelihood of winning is good. You may have chosen your course of study based on the probable availability of jobs.

You have, more than likely, used probability. In fact, you probably have an intuitive sense of probability. Probability deals with the chance of an event occurring. Whenever you weigh the odds of whether or not to do your homework or to study for an exam, you are using probability. In this chapter, you will learn how to solve probability problems using a systematic approach.

**COLLABORATIVE EXERCISE**

Your instructor will survey your class. Count the number of students in the class today.

- Raise your hand if you have any change in your pocket or purse. Record the number of raised hands.
- Raise your hand if you rode a bus within the past month. Record the number of raised hands.
- Raise your hand if you answered "yes" to BOTH of the first two questions. Record the number of raised hands.

Use the class data as estimates of the following probabilities. *P*(change) means the probability that a randomly chosen person in your class has change in his/her pocket or purse. *P*(bus) means the probability that a randomly chosen person in your class rode a bus within the last month and so on. Discuss your answers.

- Find *P*(change).

- Find $P$(bus).
- Find $P$(change AND bus). Find the probability that a randomly chosen student in your class has change in their pocket or purse and rode a bus within the last month.
- Find $P$(change|bus). Find the probability that a randomly chosen student has change given that they rode a bus within the last month. Count all the students that rode a bus. From the group of students who rode a bus, count those who have change. The probability is equal to those who have change and rode a bus divided by those who rode a bus.

## 3.1 Terminology

Probability is a measure that is associated with how certain we are of outcomes of a particular experiment or activity. An **experiment** is a planned operation carried out under controlled conditions. If the result is not predetermined, then the experiment is said to be a **chance** experiment. Flipping one fair coin twice is an example of an experiment.

A result of an experiment is called an **outcome**. The **sample space** of an experiment is the set of all possible outcomes. Three ways to represent a sample space are: to list the possible outcomes, to create a tree diagram, or to create a Venn diagram. The uppercase letter $S$ is used to denote the sample space. For example, if you flip one fair coin, $S = \{H, T\}$ where $H$ = heads and $T$ = tails are the outcomes.

An **event** is any combination of outcomes. Upper case letters like $A$ and $B$ represent events. For example, if the experiment is to flip one fair coin, event $A$ might be getting at most one head. The probability of an event $A$ is written $P(A)$.

The **probability** of any outcome is the **long-term relative frequency** of that outcome. **Probabilities are between zero and one, inclusive** (that is, zero and one and all numbers between these values). $P(A) = 0$ means the event $A$ can never happen. $P(A) = 1$ means the event $A$ always happens. $P(A) = 0.5$ means the event $A$ is equally likely to occur or not to occur. For example, if you flip one fair coin repeatedly (from 20 to 2,000 to 20,000 times) the relative frequency of heads approaches 0.5 (the probability of heads).

**Equally likely** means that each outcome of an experiment occurs with equal probability. For example, if you toss a **fair**, six-sided die, each face (1, 2, 3, 4, 5, or 6) is as likely to occur as any other face. If you toss a fair coin, a Head ($H$) and a Tail ($T$) are equally likely to occur. If you randomly guess the answer to a true/false question on an exam, you are equally likely to select a correct answer or an incorrect answer.

**To calculate the probability of an event $A$ when all outcomes in the sample space are equally likely**, count the number of outcomes for event $A$ and divide by the total number of outcomes in the sample space. For example, if you toss a fair dime and a fair nickel, the sample space is $\{HH, TH, HT, TT\}$ where $T$ = tails and $H$ = heads. The sample space has four outcomes. $A$ = getting one head. There are two outcomes that meet this condition $\{HT, TH\}$, so $P(A) = \frac{2}{4} = 0.5$.

Suppose you roll one fair six-sided die, with the numbers {1, 2, 3, 4, 5, 6} on its faces. Let event $E$ = rolling a number that is at least five. There are two outcomes {5, 6}. $P(E) = \frac{2}{6}$. If you were to roll the die only a few times, you would not be surprised if your observed results did not match the probability. If you were to roll the die a very large number of times, you would expect that, overall, $\frac{2}{6}$ of the rolls would result in an outcome of "at least five". You would not expect exactly $\frac{2}{6}$. The long-term relative frequency of obtaining this result would approach the theoretical probability of $\frac{2}{6}$ as the number of repetitions grows larger and larger.

This important characteristic of probability experiments is known as the **law of large numbers** which states that as the number of repetitions of an experiment is increased, the relative frequency obtained in the experiment tends to become closer and closer to the theoretical probability. Even though the outcomes do not happen according to any set pattern or order, overall, the long-term observed relative frequency will approach the theoretical probability. (The word **empirical** is often used instead of the word observed.)

It is important to realize that in many situations, the outcomes are not equally likely. A coin or die may be **unfair**, or **biased**. Two math professors in Europe had their statistics students test the Belgian one Euro coin and discovered that in 250 trials, a head was obtained 56% of the time and a tail was obtained 44% of the time. The data seem to show that the coin is not a fair coin; more repetitions would be helpful to draw a more accurate conclusion about such bias. Some dice may be biased. Look at the dice in a game you have at home; the spots on each face are usually small holes carved out and then painted to make the spots visible. Your dice may or may not be biased; it is possible that the outcomes may be affected by the slight weight differences due to the different numbers of holes in the faces. Gambling casinos make a lot of money depending on outcomes from rolling dice, so casino dice are made differently to eliminate bias. Casino dice

have flat faces; the holes are completely filled with paint having the same density as the material that the dice are made out of so that each face is equally likely to occur. Later we will learn techniques to use to work with probabilities for events that are not equally likely.

**"OR" Event:**
An outcome is in the event *A* OR *B* if the outcome is in *A* or is in *B* or is in both *A* and *B*.
The event A OR B can also be written as A Union B, with notation as follows:
$A \cup B$
For example, let A = {1, 2, 3, 4, 5} and B = {4, 5, 6, 7, 8}.
Then $A \cup B$ = {1, 2, 3, 4, 5, 6, 7, 8}. Notice that 4 and 5 are NOT listed twice.

**"AND" Event:**
An outcome is in the event *A* AND *B* if the outcome is in both *A* and *B* at the same time.
The event *A* AND *B* can also be written as *A* Intersection *B*, with notation as follows:
$A \cap B$
For example, let *A* and *B* be {1, 2, 3, 4, 5} and {4, 5, 6, 7, 8}, respectively.
Then $A \cap B$= {4, 5}.

The **complement of event *A*** is denoted *A*′(read "*A* prime"). *A*′consists of all outcomes that are **NOT** in *A*. Notice that *P*(*A*) + *P*(*A*′) = 1. For example, let *S* = {1, 2, 3, 4, 5, 6} and let *A* = {1, 2, 3, 4}. Then, *A*′= {5, 6}. *P*(*A*) = $\frac{4}{6}$, *P*(*A*′) = $\frac{2}{6}$, and *P*(*A*) + *P*(*A*′) = $\frac{4}{6} + \frac{2}{6} = 1$

The **conditional probability** of *A* given *B* is written *P*(*A*|*B*). *P*(*A*|*B*) is the probability that event *A* will occur given that the event *B* has already occurred. **A conditional reduces the sample space**. We calculate the probability of *A* from the reduced sample space *B*. The formula to calculate *P*(*A*|*B*) is $P(A|B) = \frac{P(A \text{ AND } B)}{P(B)}$ where *P*(*B*) is greater than zero.

For example, suppose we toss one fair, six-sided die. The sample space *S* = {1, 2, 3, 4, 5, 6}. Let *A* = face is 2 or 3 and *B* = face is even (2, 4, 6). To calculate *P*(*A*|*B*), we count the number of outcomes 2 or 3 in the sample space *B* = {2, 4, 6}. Then we divide that by the number of outcomes *B* (rather than *S*).

We get the same result by using the formula. Remember that *S* has six outcomes.

$$P(A|B) = \frac{P(A \text{ AND } B)}{P(B)} = \frac{\frac{(\text{the number of outcomes that are 2 or 3 and even in } S)}{6}}{\frac{(\text{the number of outcomes that are even in } S)}{6}} = \frac{\frac{1}{6}}{\frac{3}{6}} = \frac{1}{3}$$

**Understanding Terminology and Symbols**
It is important to read each problem carefully to think about and understand what the events are. Understanding the wording is the first very important step in solving probability problems. Reread the problem several times if necessary. Clearly identify the event of interest. Determine whether there is a condition stated in the wording that would indicate that the probability is conditional; carefully identify the condition, if any.

**EXAMPLE 3.1**

**Problem**
The sample space *S* is the whole numbers starting at one and less than 20.

a. *S* = _____
   Let event *A* = the even numbers and event *B* = numbers greater than 13.
b. *A* = _____, *B* = _____
c. *P*(*A*) = _____, *P*(*B*) = _____
d. *A* AND *B* = _____, *A* OR *B* = _____
e. *P*(*A* AND *B*) = _____, *P*(*A* OR *B*) = _____
f. *A*′= _____, *P*(*A*′) = _____
g. *P*(*A*) + *P*(*A*′) = _____
h. *P*(*A*|*B*) = _____, *P*(*B*|*A*) = _____; are the probabilities equal?

**Solution**
a. *S* = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19}

b. $A$ = {2, 4, 6, 8, 10, 12, 14, 16, 18}, $B$ = {14, 15, 16, 17, 18, 19}
c. $P(A) = \frac{9}{19}$, $P(B) = \frac{6}{19}$
d. $A$ AND $B$ = {14,16,18}, $A$ OR $B$ = {2, 4, 6, 8, 10, 12, 14, 15, 16, 17, 18, 19}
e. $P(A \text{ AND } B) = \frac{3}{19}$, $P(A \text{ OR } B) = \frac{12}{19}$
f. $A'$ = 1, 3, 5, 7, 9, 11, 13, 15, 17, 19; $P(A') = \frac{10}{19}$
g. $P(A) + P(A') = 1$ ($\frac{9}{19} + \frac{10}{19} = 1$)
h. $P(A|B) = \frac{P(A \text{ AND } B)}{P(B)} = \frac{3}{6}$, $P(B|A) = \frac{P(A \text{ AND } B)}{P(A)} = \frac{3}{9}$, No

**TRY IT 3.1**

The sample space $S$ is all the ordered pairs of two whole numbers, the first from one to three and the second from one to four (Example: (1, 4)).

a. $S$ = _____

Let event $A$ = the sum is even and event $B$ = the first number is prime.
b. $A$ = _____, $B$ = _____
c. $P(A)$ = _____, $P(B)$ = _____
d. $A$ AND $B$ = _____, $A$ OR $B$ = _____
e. $P(A \text{ AND } B)$ = _____, $P(A \text{ OR } B)$ = _____
f. $B'$ = _____, $P(B')$ = _____
g. $P(A) + P(A')$ = _____
h. $P(A|B)$ = _____, $P(B|A)$ = _____; are the probabilities equal?

**EXAMPLE 3.2**

**Problem**

A fair, six-sided die is rolled. Describe the sample space $S$, identify each of the following events with a subset of $S$ and compute its probability (an outcome is the number of dots that show up).

a. Event $T$ = the outcome is two.
b. Event $A$ = the outcome is an even number.
c. Event $B$ = the outcome is less than four.
d. The complement of $A$.
e. $A$ GIVEN $B$
f. $B$ GIVEN $A$
g. $A$ AND $B$
h. $A$ OR $B$
i. $A$ OR $B'$
j. Event $N$ = the outcome is a prime number.
k. Event $I$ = the outcome is seven.

**Solution**

a. $T$ = {2}, $P(T) = \frac{1}{6}$
b. $A$ = {2, 4, 6}, $P(A) = \frac{1}{2}$
c. $B$ = {1, 2, 3}, $P(B) = \frac{1}{2}$
d. $A'$ = {1, 3, 5}, $P(A') = \frac{1}{2}$
e. $A|B$ = {2}, $P(A|B) = \frac{1}{3}$
f. $B|A$ = {2}, $P(B|A) = \frac{1}{3}$
g. $A$ AND $B$ = {2}, $P(A \text{ AND } B) = \frac{1}{6}$

h. $A$ OR $B$ = {1, 2, 3, 4, 6}, $P(A \text{ OR } B) = \frac{5}{6}$
i. $A$ OR $B' = \{2, 4, 5, 6\}$, $P(A \text{ OR } B') = \frac{2}{3}$
j. $N = \{2, 3, 5\}$, $P(N) = \frac{1}{2}$
k. A six-sided die does not have seven dots. $P(7) = 0$.

**TRY IT 3.2**

A number is randomly selected between 1 to 10. Describe the sample space $S$, identify each of the following events with a subset of S, and compute its probability (an outcome is the number selected).

a. Event $F$ = the outcome is 5.
b. Event $A$ = the number is more than 6.
c. Event $B$ = the number is odd.
d. The complement of $B$.
e. $A$ GIVEN $B$.
f. $B$ GIVEN $A$.
g. $A$ OR $B$.
h. $A'$ OR $B$.
i. Event $P$ = the outcome is a composite number.
j. Event $M$ = the number is a multiple of 3.

**EXAMPLE 3.3**

Table 3.1 describes the distribution of a random sample $S$ of 100 individuals, organized by sex assigned at birth and whether they are right- or left-handed.

| | Right-handed | Left-handed |
|---|---|---|
| Males | 43 | 9 |
| Females | 44 | 4 |

**Table 3.1**

**Problem**

Let's denote the events $M$ = the subject is male, $F$ = the subject is female, $R$ = the subject is right-handed, $L$ = the subject is left-handed. Compute the following probabilities:

a. $P(M)$
b. $P(F)$
c. $P(R)$
d. $P(L)$
e. $P(M \text{ AND } R)$
f. $P(F \text{ AND } L)$
g. $P(M \text{ OR } F)$
h. $P(M \text{ OR } R)$
i. $P(F \text{ OR } L)$
j. $P(M')$
k. $P(R \mid M)$
l. $P(F \mid L)$
m. $P(L \mid F)$

**Solution**

a. $P(M) = 0.52$
b. $P(F) = 0.48$
c. $P(R) = 0.87$
d. $P(L) = 0.13$
e. $P(M \text{ AND } R) = 0.43$
f. $P(F \text{ AND } L) = 0.04$
g. $P(M \text{ OR } F) = 1$
h. $P(M \text{ OR } R) = 0.96$
i. $P(F \text{ OR } L) = 0.57$
j. $P(M') = 0.48$
k. $P(R \mid M) = 0.8269$ (rounded to four decimal places)
l. $P(F \mid L) = 0.3077$ (rounded to four decimal places)
m. $P(L \mid F) = 0.0833$

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**TRY IT 3.3**

The table describes the distribution of a random sample S of 100 individuals, organized by gender and whether they like tea or coffee.

| | Tea | Coffee |
|---|---|---|
| **Men** | 22 | 26 |
| **Women** | 16 | 36 |

**Table 3.2**

Let's denote the events $M$ = the subject is a man, $W$ = the subject is a woman, $T$ = the subject likes tea, and $C$ = the subject likes coffee. Compute the following probabilities:

a. $P(M)$
b. $P(W)$
c. $P(T)$
d. $P(C)$
e. $P(M \text{ AND } T)$
f. $P(W \text{ and } C)$
g. $P(M \text{ OR } W)$
h. $P(M \text{ OR } T)$
i. $P(W \text{ OR } C)$
j. $P(M')$
k. $P(T \mid M)$
l. $P(W \mid C)$
m. $P(C \mid W)$

# 3.2 Independent and Mutually Exclusive Events

Independent and mutually exclusive do **not** mean the same thing.

## Independent Events

Two events are independent if the following are true:

- $P(A \mid B) = P(A)$
- $P(B \mid A) = P(B)$

- $P(A \text{ AND } B) = P(A)P(B)$

Two events $A$ and $B$ are **independent events** if the knowledge that one occurred does not affect the chance the other occurs. For example, the outcomes of two roles of a fair die are independent events. The outcome of the first roll does not change the probability for the outcome of the second roll. To show two events are independent, you must show **only one** of the above conditions. If two events are NOT independent, then we say that they are **dependent events**.

Sampling may be done **with replacement** or **without replacement**.

- **With replacement**: If each member of a population is replaced after it is picked, then that member has the possibility of being chosen more than once. When sampling is done with replacement, then events are considered to be independent, meaning the result of the first pick will not change the probabilities for the second pick.
- **Without replacement**: When sampling is done without replacement, each member of a population may be chosen only once. In this case, the probabilities for the second pick are affected by the result of the first pick. The events are considered to be dependent or not independent.

If it is not known whether $A$ and $B$ are independent or dependent, **assume they are dependent until you can show otherwise**.

**EXAMPLE 3.4**

You have a fair, well-shuffled deck of 52 cards. It consists of four suits. The suits are clubs, diamonds, hearts and spades. There are 13 cards in each suit consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, $J$ (jack), $Q$ (queen), $K$ (king) of that suit.

a. Sampling with replacement:
Suppose you pick three cards with replacement. The first card you pick out of the 52 cards is the $Q$ of spades. You put this card back, reshuffle the cards and pick a second card from the 52-card deck. It is the ten of clubs. You put this card back, reshuffle the cards and pick a third card from the 52-card deck. This time, the card is the $Q$ of spades again. Your picks are {$Q$ of spades, ten of clubs, $Q$ of spades}. You have picked the $Q$ of spades twice. You pick each card from the 52-card deck.

b. Sampling without replacement:
Suppose you pick three cards without replacement. The first card you pick out of the 52 cards is the $K$ of hearts. You put this card aside and pick the second card from the 51 cards remaining in the deck. It is the three of diamonds. You put this card aside and pick the third card from the remaining 50 cards in the deck. The third card is the $J$ of spades. Your picks are {$K$ of hearts, three of diamonds, $J$ of spades}. Because you have picked the cards without replacement, you cannot pick the same card twice.

**TRY IT 3.4**

You have a fair, well-shuffled deck of 52 cards. It consists of four suits. The suits are clubs, diamonds, hearts and spades. There are 13 cards in each suit consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, $J$ (jack), $Q$ (queen), $K$ (king) of that suit. Three cards are picked at random.

a. Suppose you know that the picked cards are $Q$ of spades, $K$ of hearts and $Q$ of spades. Can you decide if the sampling was with or without replacement?
b. Suppose you know that the picked cards are $Q$ of spades, $K$ of hearts, and $J$ of spades. Can you decide if the sampling was with or without replacement?

**EXAMPLE 3.5**

**Problem**
You have a fair, well-shuffled deck of 52 cards. It consists of four suits. The suits are clubs, diamonds, hearts, and spades. There are 13 cards in each suit consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, $J$ (jack), $Q$ (queen), and $K$ (king) of that suit. $S$ = spades, $H$ = Hearts, $D$ = Diamonds, $C$ = Clubs.

a. Suppose you pick four cards, but do not put any cards back into the deck. Your cards are $QS$, $1D$, $1C$, $QD$.

b. Suppose you pick four cards and put each card back before you pick the next card. Your cards are *KH*, 7*D*, 6*D*, *KH*.

Which of a. or b. did you sample with replacement and which did you sample without replacement?

**Solution**

a. Without replacement; b. With replacement

**TRY IT 3.5**

You have a fair, well-shuffled deck of 52 cards. It consists of four suits. The suits are clubs, diamonds, hearts, and spades. There are 13 cards in each suit consisting of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, *J* (jack), *Q* (queen), and *K* (king) of that suit. *S* = spades, *H* = Hearts, *D* = Diamonds, *C* = Clubs. Suppose that you sample four cards without replacement. Which of the following outcomes are possible? Answer the same question for sampling with replacement.

a. *QS*, 1*D*, 1*C*, *QD*
b. *KH*, 7*D*, 6*D*, *KH*
c. *QS*, 7*D*, 6*D*, *KS*

## Mutually Exclusive Events

*A* and *B* are **mutually exclusive** events if they cannot occur at the same time. This means that *A* and *B* do not share any outcomes and *P*(*A* AND *B*) = 0.

For example, suppose the sample space *S* = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Let *A* = {1, 2, 3, 4, 5}, *B* = {4, 5, 6, 7, 8}, and *C* = {7, 9}. *A* AND *B* = {4, 5}. *P*(*A* AND *B*) = $\frac{2}{10}$ and is not equal to zero. Therefore, *A* and *B* are not mutually exclusive. *A* and *C* do not have any numbers in common so *P*(*A* AND *C*) = 0. Therefore, *A* and *C* are mutually exclusive.

If it is not known whether *A* and *B* are mutually exclusive, **assume they are not until you can show otherwise**. The following examples illustrate these definitions and terms.

**EXAMPLE 3.6**

Flip two fair coins. (This is an experiment.)

The sample space is {*HH*, *HT*, *TH*, *TT*} where *T* = tails and *H* = heads. The outcomes are *HH*, *HT*, *TH*, and *TT*. The outcomes HT and TH are different. The *HT* means that the first coin showed heads and the second coin showed tails. The *TH* means that the first coin showed tails and the second coin showed heads.

- Let *A* = the event of getting **at most one tail**. (At most one tail means zero or one tail.) Then *A* can be written as {*HH*, *HT*, *TH*}. The outcome *HH* shows zero tails. *HT* and *TH* each show one tail.
- Let *B* = the event of getting all tails. *B* can be written as {*TT*}. *B* is the **complement** of *A*, so *B* = *A*′. Also, *P*(*A*) + *P*(*B*) = *P*(*A*) + *P*(*A*′) = 1.
- The probabilities for *A* and for *B* are *P*(*A*) = $\frac{3}{4}$ and *P*(*B*) = $\frac{1}{4}$.
- Let *C* = the event of getting all heads. *C* = {*HH*}. Since *B* = {*TT*}, *P*(*B* AND *C*) = 0. *B* and *C* are mutually exclusive. (*B* and *C* have no members in common because you cannot have all tails and all heads at the same time.)
- Let *D* = event of getting **more than one** tail. *D* = {*TT*}. *P*(*D*) = $\frac{1}{4}$
- Let *E* = event of getting a head on the first roll. (This implies you can get either a head or tail on the second roll.) *E* = {*HT*, *HH*}. *P*(*E*) = $\frac{2}{4}$
- Find the probability of getting **at least one** (one or two) tail in two flips. Let *F* = event of getting at least one tail in two flips. *F* = {*HT*, *TH*, *TT*}. *P*(*F*) = $\frac{3}{4}$

**TRY IT 3.6**

Draw two cards from a standard 52-card deck with replacement. Find the probability of getting at least one black card.

**EXAMPLE 3.7**

**Problem**

Flip two fair coins. Find the probabilities of the events.

a. Let $F$ = the event of getting at most one tail (zero or one tail).
b. Let $G$ = the event of getting two faces that are the same.
c. Let $H$ = the event of getting a head on the first flip followed by a head or tail on the second flip.
d. Are $F$ and $G$ mutually exclusive?
e. Let $J$ = the event of getting all tails. Are $J$ and $H$ mutually exclusive?

**Solution**

Look at the sample space in Example 3.6.

a. Zero (0) or one (1) tails occur when the outcomes $HH$, $TH$, $HT$ show up. $P(F) = \frac{3}{4}$
b. Two faces are the same if $HH$ or $TT$ show up. $P(G) = \frac{2}{4}$
c. A head on the first flip followed by a head or tail on the second flip occurs when $HH$ or $HT$ show up. $P(H) = \frac{2}{4}$
d. $F$ and $G$ share $HH$ so $P(F \text{ AND } G)$ is not equal to zero (0). $F$ and $G$ are not mutually exclusive.
e. Getting all tails occurs when tails shows up on both coins ($TT$). $H$'s outcomes are $HH$ and $HT$.

$J$ and $H$ have nothing in common so $P(J \text{ AND } H) = 0$. $J$ and $H$ are mutually exclusive.

**TRY IT 3.7**

A box has two balls, one white and one red. We select one ball, put it back in the box, and select a second ball (sampling with replacement). Find the probability of the following events:

a. Let $F$ = the event of getting the white ball twice.
b. Let $G$ = the event of getting two balls of different colors.
c. Let $H$ = the event of getting white on the first pick.
d. Are $F$ and $G$ mutually exclusive?
e. Are $G$ and $H$ mutually exclusive?

**EXAMPLE 3.8**

Roll one fair, six-sided die. The sample space is {1, 2, 3, 4, 5, 6}. Let event $A$ = a face is odd. Then $A$ = {1, 3, 5}. Let event $B$ = a face is even. Then $B$ = {2, 4, 6}.

- Find the complement of $A$, $A'$. The complement of $A$, $A'$, is $B$ because $A$ and $B$ together make up the sample space. $P(A) + P(B) = P(A) + P(A') = 1$. Also, $P(A) = \frac{3}{6}$ and $P(B) = \frac{3}{6}$.
- Let event $C$ = odd faces larger than two. Then $C$ = {3, 5}. Let event $D$ = all even faces smaller than five. Then $D$ = {2, 4}. $P(C \text{ AND } D) = 0$ because you cannot have an odd and even face at the same time. Therefore, $C$ and $D$ are mutually exclusive events.
- Let event $E$ = all faces less than five. $E$ = {1, 2, 3, 4}.

**Problem**

Are $C$ and $E$ mutually exclusive events? (Answer yes or no.) Why or why not?

**Solution**

No. $C$ = {3, 5} and $E$ = {1, 2, 3, 4}. $P(C \text{ AND } E) = \frac{1}{6}$. To be mutually exclusive, $P(C \text{ AND } E)$ must be zero.

- Find $P(C|A)$. This is a conditional probability. Recall that the event $C$ is {3, 5} and event $A$ is {1, 3, 5}. To find $P(C|A)$, find the probability of $C$ using the sample space $A$. You have reduced the sample space from the original sample space {1, 2, 3, 4, 5, 6} to {1, 3, 5}. So, $P(C|A) = \frac{2}{3}$.

**TRY IT 3.8**

Let event $A$ = learning Spanish. Let event $B$ = learning German. Then $A$ AND $B$ = learning Spanish and German. Suppose $P(A) = 0.4$ and $P(B) = 0.2$. $P(A \text{ AND } B) = 0.08$. Are events $A$ and $B$ independent? Hint: You must show ONE of the following:

- $P(A|B) = P(A)$
- $P(B|A) = P(B)$
- $P(A \text{ AND } B) = P(A)P(B)$

**EXAMPLE 3.9**

Let event $G$ = taking a math class. Let event $H$ = taking a science class. Then, $G$ AND $H$ = taking a math class and a science class. Suppose $P(G) = 0.6$, $P(H) = 0.5$, and $P(G \text{ AND } H) = 0.3$. Are $G$ and $H$ independent?

If $G$ and $H$ are independent, then you must show **ONE** of the following:

- $P(G|H) = P(G)$
- $P(H|G) = P(H)$
- $P(G \text{ AND } H) = P(G)P(H)$

**NOTE**

**The choice you make depends on the information you have.** You could choose any of the methods here because you have the necessary information.

**Problem**

a. Show that $P(G|H) = P(G)$.

**Solution**

$P(G|H) = \frac{P(G \text{ AND } H)}{P(H)} = \frac{0.3}{0.5} = 0.6 = P(G)$

**Problem**

b. Show $P(G \text{ AND } H) = P(G)P(H)$.

**Solution**

$P(G)P(H) = (0.6)(0.5) = 0.3 = P(G \text{ AND } H)$

Since $G$ and $H$ are independent, knowing that a person is taking a science class does not change the chance that they are taking a math class. If the two events had not been independent (that is, they are dependent) then knowing that a person is taking a science class would change the chance they are taking math. For practice, show that $P(H|G) = P(H)$ to show that $G$ and $H$ are independent events.

**TRY IT 3.9**

In a bag, there are six red marbles and four green marbles. The red marbles are marked with the numbers 1, 2, 3, 4, 5, and 6. The green marbles are marked with the numbers 1, 2, 3, and 4.

- $R$ = a red marble
- $G$ = a green marble
- $O$ = an odd-numbered marble
- The sample space is $S = \{R1, R2, R3, R4, R5, R6, G1, G2, G3, G4\}$.

$S$ has ten outcomes. What is $P(G \text{ AND } O)$?

**EXAMPLE 3.10**

**Problem**

Let event $C$ = taking an English class. Let event $D$ = taking a speech class.

Suppose $P(C) = 0.75$, $P(D) = 0.3$, $P(C|D) = 0.75$ and $P(C \text{ AND } D) = 0.225$.

Justify your answers to the following questions numerically.

a. Are $C$ and $D$ independent?
b. Are $C$ and $D$ mutually exclusive?
c. What is $P(D|C)$?

**Solution**

a. Yes, because $P(C|D) = P(C)$.
b. No, because $P(C \text{ AND } D)$ is not equal to zero.
c. $P(D|C) = \frac{P(C \text{ AND } D)}{P(C)} = \frac{0.225}{0.75} = 0.3$

**TRY IT 3.10**

A student goes to the library. Let events $B$ = the student checks out a book and $D$ = the student checks out a DVD. Suppose that $P(B) = 0.40$, $P(D) = 0.30$ and $P(B \text{ AND } D) = 0.20$.

a. Find $P(B|D)$.
b. Find $P(D|B)$.
c. Are $B$ and $D$ independent?
d. Are $B$ and $D$ mutually exclusive?

**EXAMPLE 3.11**

In a box there are three red cards and five blue cards. The red cards are marked with the numbers 1, 2, and 3, and the blue cards are marked with the numbers 1, 2, 3, 4, and 5. The cards are well-shuffled. You reach into the box (you cannot see into it) and draw one card.

Let $R$ = red card is drawn, $B$ = blue card is drawn, $E$ = even-numbered card is drawn.

The sample space $S = R1, R2, R3, B1, B2, B3, B4, B5$. $S$ has eight outcomes.

- $P(R) = \frac{3}{8}$. $P(B) = \frac{5}{8}$. $P(R \text{ AND } B) = 0$. (You cannot draw one card that is both red and blue.)
- $P(E) = \frac{3}{8}$. (There are three even-numbered cards, $R2$, $B2$, and $B4$.)
- $P(E|B) = \frac{2}{5}$. (There are five blue cards: $B1$, $B2$, $B3$, $B4$, and $B5$. Out of the blue cards, there are two even cards; $B2$ and $B4$.)
- $P(B|E) = \frac{2}{3}$. (There are three even-numbered cards: $R2$, $B2$, and $B4$. Out of the even-numbered cards, to are blue; $B2$ and $B4$.)
- The events $R$ and $B$ are mutually exclusive because $P(R \text{ AND } B) = 0$.
- Let $G$ = card with a number greater than 3. $G = \{B4, B5\}$. $P(G) = \frac{2}{8}$. Let $H$ = blue card numbered between one and four, inclusive. $H = \{B1, B2, B3, B4\}$. $P(G|H) = \frac{1}{4}$. (The only card in $H$ that has a number greater than three is $B4$.) Since $\frac{2}{8} = \frac{1}{4}$, $P(G) = P(G|H)$, which means that $G$ and $H$ are independent.

**TRY IT 3.11**

In a basketball arena,

- 70% of the fans are rooting for the home team.
- 25% of the fans are wearing blue.
- 20% of the fans are wearing blue and are rooting for the away team.
- Of the fans rooting for the away team, 67% are wearing blue.

Let $A$ be the event that a fan is rooting for the away team.
Let $B$ be the event that a fan is wearing blue.
Are the events of rooting for the away team and wearing blue independent? Are they mutually exclusive?

**EXAMPLE 3.12**

**Problem**

In a particular college class, 60% of the students are women. Fifty percent of all students in the class have long hair. Forty-five percent of the students are women and have long hair. Of the women students, 75% have long hair. Let $W$ be the event that a student is a women. Let $L$ be the event that a student has long hair. One student is picked randomly. Are the events of being a woman and having long hair independent?

- The following probabilities are given in this example:
- $P(W) = 0.60$; $P(L) = 0.50$
- $P(W \text{ AND } L) = 0.45$
- $P(L \mid W) = 0.75$

**NOTE**

**The choice you make depends on the information you have.** You could use the first or last condition on the list for this example. You do not know $P(W \mid L)$ yet, so you cannot use the second condition.

**Solution**

>Check whether $P(W \text{ AND } L) = P(W)P(L)$. We are given that $P(W \text{ AND } L) = 0.45$, but $P(W)P(L) = (0.60)(0.50) = 0.30$. The events of being a woman and having long hair are not independent because $P(W \text{ AND } L)$ does not equal $P(W)P(L)$.

Check whether $P(L \mid W)$ equals $P(L)$. We are given that $P(L \mid W) = 0.75$, but $P(L) = 0.50$; they are not equal. The events of being a woman and having long hair are not independent.

The events of being a woman and having long hair are not independent; knowing that a student is a woman changes the probability that a student has long hair.

**TRY IT 3.12**

Mark is deciding which route to take to work. His choices are $I$ = the Interstate and $F$ = Fifth Street.

- $P(I) = 0.44$ and $P(F) = 0.56$
- $P(I \text{ AND } F) = 0$ because Mark will take only one route to work.

What is the probability of $P(I \text{ OR } F)$?

**EXAMPLE 3.13**

**Problem**

a. Toss one fair coin (the coin has two sides, $H$ and $T$). The outcomes are _____. Count the outcomes. There are ____ outcomes.
b. Toss one fair, six-sided die (the die has 1, 2, 3, 4, 5 or 6 dots on a side). The outcomes are _____. Count the outcomes. There are ___ outcomes.